

Payout Retention Signal and the Cross Section of Stock Returns

I. M. Harking

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Abstract

This paper studies the asset pricing implications of Payout Retention Signal (PRS), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by [Novy-Marx and Velikov \(2023\)](#). A value-weighted long/short trading strategy based on PRS achieves an annualized gross (net) Sharpe ratio of 0.45 (0.39), and monthly average abnormal gross (net) return relative to the [Fama and French \(2015\)](#) five-factor model plus a momentum factor of 26 (23) bps/month with a t-statistic of 3.22 (2.94), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Growth in book equity, Net Payout Yield, Share issuance (1 year), Share issuance (5 year), Change in equity to assets, Asset growth) is 18 bps/month with a t-statistic of 2.44.

1 Introduction

Financial markets efficiently incorporate information about firms’ productive capacity and investment opportunities into asset prices, yet significant cross-sectional variation in expected returns persists even after controlling for standard risk factors. This variation reflects differences in firms’ exposure to aggregate productivity shocks and their ability to adjust capital in response to changing investment opportunities. We examine how firms’ payout retention decisions signal their marginal productivity of capital and exposure to investment frictions, providing insight into the production-based determinants of expected returns. The Payout Retention Signal (PRS) captures the fraction of earnings retained after distributions to shareholders, reflecting managers’ assessment of internal investment opportunities relative to the cost of capital. Firms with higher retention rates signal greater exposure to productivity shocks and face higher adjustment costs when reallocating capital, suggesting these firms should command higher risk premia in equilibrium.

In a production-based asset pricing framework, firms’ payout policies directly reflect the marginal productivity of capital and the severity of investment frictions they face [Cochrane \(1991\)](#). When firms retain earnings rather than distribute them to shareholders, they signal that internal investment opportunities exceed the market’s required return, consistent with binding financing constraints or high adjustment costs that make external capital costly [Zhang \(2005\)](#). These retention decisions map directly to firms’ exposure to aggregate productivity shocks through the investment channel—firms with volatile investment opportunities require precautionary cash holdings to avoid costly external financing when positive shocks arrive [Berk et al. \(2004\)](#). The cross-sectional variation in payout retention therefore proxies for heterogeneity in firms’ conditional risk premia. Firms retaining more earnings face steeper marginal costs of investment adjustment, making their cash flows more sensitive to aggregate productivity fluctuations [Cooper and Haltiwanger \(2006\)](#). This

sensitivity manifests through the operating leverage channel: high-retention firms have committed to fixed investment programs that amplify their exposure to systematic risk [Carlson et al. \(2004\)](#). Moreover, the decision to retain earnings rather than pay dividends reveals managers' private information about future productivity, with retention signaling exposure to persistent productivity shocks that require internal funding [Liu et al. \(2009\)](#). The production-based view predicts that firms with higher payout retention ratios should earn higher expected returns as compensation for their greater exposure to systematic investment shocks.

The Payout Retention Signal generates economically and statistically significant return predictability in the cross-section of U.S. equities. A value-weighted long-short portfolio that buys stocks with high PRS and sells stocks with low PRS earns an average monthly return of 28 basis points with a t-statistic of 3.51, corresponding to an annualized Sharpe ratio of 0.45. The strategy's alpha remains robust across standard factor models, with the six-factor alpha equaling 26 basis points per month (t-statistic = 3.22). After accounting for transaction costs using composite bid-ask spreads, the strategy maintains a net Sharpe ratio of 0.39 and delivers a net six-factor alpha of 23 basis points per month with a t-statistic of 2.94. The predictive power of PRS concentrates in firms facing the greatest investment frictions—the strategy loads significantly on the CMA factor with a coefficient of 0.29 (t-statistic = 5.26), confirming that high-PRS firms exhibit conservative investment patterns consistent with binding adjustment costs. Importantly, the PRS effect persists among the largest stocks where implementation is most feasible: within the top NYSE size quintile, the strategy generates 27 basis points per month (t-statistic = 2.93) with minimal degradation in risk-adjusted performance. When we control for the six most closely related anomalies from the factor zoo plus the six Fama-French factors, PRS retains an alpha of 18 basis points per month with a t-statistic of 2.44, demonstrating that the signal captures unique information about firms' exposure to productivity

shocks not subsumed by existing predictors.

This paper contributes to the production-based asset pricing literature by establishing payout retention as a direct measure of firms' exposure to investment frictions and productivity risk. While [Fama and French \(2015\)](#) document that conservative investment predicts returns through the CMA factor, we show that payout retention provides incremental predictive power by capturing firms' forward-looking assessment of productivity shocks and adjustment costs. Our findings extend [Hou et al. \(2015\)](#) by demonstrating that retention decisions reflect not just current profitability and investment, but also managers' private information about future productivity dynamics that standard accounting variables fail to capture. Unlike the mechanical asset growth effect documented in [Cooper et al. \(2008\)](#), PRS specifically isolates the component of internal financing that signals exposure to costly external finance and investment irreversibility. We advance the methodology of [Liu et al. \(2009\)](#) by showing that a simple payout-based measure effectively proxies for the shadow value of capital without requiring structural estimation. The economic magnitude and statistical robustness of PRS across size groups addresses concerns raised by [Novy-Marx and Velikov \(2016a\)](#) about the implementability of anomalies, as our strategy generates significant net returns even among large-cap stocks with low transaction costs. By linking payout policy directly to production-based risk factors, we provide new evidence that cross-sectional return predictability reflects rational compensation for exposure to systematic investment shocks rather than behavioral mispricing. These results have important implications for portfolio construction and risk management, as PRS offers a parsimonious signal that captures firms' endogenous risk exposure through their optimal response to productivity shocks and capital market frictions.

2 Data

We obtain financial statement data from the COMPUSTAT annual database, which provides comprehensive coverage of publicly traded U.S. companies. Our sample includes all firms with available data for the requisite variables, spanning several decades of financial reporting. We focus on annual observations to align with the frequency of financial statement reporting and to ensure consistency in our signal construction. Payout Retention Signal relies on two key COMPUSTAT variables. Common stock (CSTK) represents the dollar amount of funds received from the issuance of common stock during the fiscal year, capturing new equity raised through primary offerings, employee stock option exercises, and other common equity issuances. Dividends (DVT) measures the total dollar amount of cash dividends declared on common stock during the fiscal year, representing the firm's cash distributions to common shareholders. Both variables are reported in millions of dollars and reflect flows that occur during the fiscal year rather than point-in-time balance sheet values. We construct the Payout Retention Signal by computing the year-over-year change in common stock issuance, scaled by lagged dividends: $(CSTK(t) - CSTK(t-1)) / DVT(t-1)$. This ratio captures the change in equity issuance activity relative to the firm's dividend payout capacity, providing insight into how firms balance external equity financing with their dividend distribution policies. We calculate the signal using fiscal year-end values and require non-missing values for common stock in consecutive years and non-zero, positive dividends in the lagged period to ensure meaningful scaling. Firms with zero or negative lagged dividends are excluded from the analysis as the scaling becomes undefined or economically uninterpretable. The resulting signal measures the intensity of changes in equity issuance relative to the firm's established payout level, with higher values indicating increased reliance on external equity financing relative to dividend capacity.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the PRS signal. Panel A plots the time-series of the mean, median, and interquartile range for PRS. On average, the cross-sectional mean (median) PRS is -1.46 (-0.01) over the 1963 to 2024 sample, where the starting date is determined by the availability of the input PRS data. The signal’s interquartile range spans -0.37 to 0.00. Panel B of Figure 1 plots the time-series of the coverage of the PRS signal for the CRSP universe. On average, the PRS signal is available for 3.46% of CRSP names, which on average make up 6.53% of total market capitalization.

4 Does PRS predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on PRS using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high PRS portfolio and sells the low PRS portfolio. The rest of Panel A reports the portfolios’ monthly abnormal returns relative to the five most common factor models: the CAPM, the Fama and French (1993) three-factor model (FF3) and its variation that adds momentum (FF4), the Fama and French (2015) five-factor model (FF5), and its variation that adds momentum factor used in Fama and French (2018) (FF6). The table shows that the long/short PRS strategy earns an average return of 0.28% per month with a t-statistic of 3.51. The annualized Sharpe ratio of the strategy is 0.45. The alphas range from 0.26% to 0.32% per month and have t-statistics exceeding 3.22 everywhere. The lowest alpha is with respect to the FF5 factor model.

Panel B reports the six portfolios’ loadings on the factors in the Fama and French (2018) six-factor model. The long/short strategy’s most significant loading is 0.29,

with a t-statistic of 5.26 on the CMA factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in \$ millions) of the stocks they hold. In an average month, the five portfolios have at least 317 stocks and an average market capitalization of at least \$1,150 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor’s achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different portfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using cap breakpoints and value-weighted portfolios, and equals 23 bps/month with a t-statistics of 3.03. Out of the twenty-five alphas reported in Panel A, the t-statistics for twenty-five exceed two, and for twenty-four exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of [Novy-Marx and Velikov \(2016b\)](#). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from [Chen and Velikov \(2022\)](#). The net average returns

reported in the first column range between 19-31bps/month. The lowest return, (19 bps/month), is achieved from the quintile sort using NYSE breakpoints and equal-weighted portfolios, and has an associated t-statistic of 3.31. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the PRS trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty-five cases, and significantly expands the achievable frontier in twenty-five cases.

Table 3 provides direct tests for the role size plays in the PRS strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and PRS, as well as average returns and alphas for long/short trading PRS strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80th NYSE percentile), the PRS strategy achieves an average return of 27 bps/month with a t-statistic of 2.93. Among these large cap stocks, the alphas for the PRS strategy relative to the five most common factor models range from 27 to 30 bps/month with t-statistics between 2.90 and 3.29.

5 How does PRS perform relative to the zoo?

Figure 2 puts the performance of PRS in context, showing the long/short strategy performance relative to other strategies in the “factor zoo.” It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.¹ The vertical red line shows where the Sharpe ratio for the PRS strategy falls in the distribution. The PRS strategy’s gross (net) Sharpe ratio of 0.45 (0.39) is greater than 85% (95%) of anomaly Sharpe ratios,

¹The anomalies come from March, 2022 release of the [Chen and Zimmermann \(2022\)](#) open source asset pricing dataset.

respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the PRS strategy (red line).² Ignoring trading costs, a \$1 invested in the PRS strategy would have yielded \$5.41 which ranks the PRS strategy in the top 5% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the PRS strategy would have yielded \$3.92 which ranks the PRS strategy in the top 3% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and Novy-Marx and Velikov (2016b) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the PRS relative to those. Panel A shows that the PRS strategy gross alphas fall between the 61 and 74 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 196306 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The PRS strategy has a positive net generalized alpha for five out of the five factor models. In these cases PRS ranks between the 79 and 88 percentiles in terms of how much it could have expanded the achievable investment frontier.

²The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in Detzel et al. (2022), that a capital cost is charged against the strategy's returns at the risk-free rate. This excess return corresponds more closely to the strategy's economic profitability.

6 Does PRS add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of PRS with 205 filtered anomaly signals.³ Figure 6 also shows an agglomerative hierarchical cluster plot using Ward’s minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price PRS or at least to weaken the power PRS has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of PRS conditioning on each of the 205 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{PRS} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{PRS}PRS_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 205 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{PRS,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 205 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 205 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on PRS. Stocks are finally grouped into five PRS portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted

³When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

PRS trading strategies conditioned on each of the 205 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on PRS and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the PRS signal in these Fama-MacBeth regressions exceed 1.74, with the minimum t-statistic occurring when controlling for Net Payout Yield. Controlling for all six closely related anomalies, the t-statistic on PRS is 1.03.

Similarly, Table 5 reports results from spanning tests that regress returns to the PRS strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the PRS strategy earns alphas that range from 18-29bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 2.32, which is achieved when controlling for Net Payout Yield. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the PRS trading strategy achieves an alpha of 18bps/month with a t-statistic of 2.44.

7 Does PRS add relative to the whole zoo?

Finally, we can ask how much adding PRS to the entire factor zoo could improve investment performance. Figure 8 plots the growth of \$1 invested in trading strategies that combine multiple anomalies following Chen and Velikov (2022). The combinations use either the 154 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 154 anomalies augmented with the PRS signal.⁴ We consider one different methods for combining signals.

⁴We filter the 207 Chen and Zimmermann (2022) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which PRS is available.

Panel A shows results using “Average rank” as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, \$1 investment in the 154-anomaly combination strategy grows to \$2560.30, while \$1 investment in the combination strategy that includes PRS grows to \$2783.47.

8 Conclusion

This study provides robust evidence that the Payout Retention Signal (PRS) represents a economically meaningful and statistically significant predictor of cross-sectional stock returns. Our comprehensive analysis, following the rigorous testing protocol of Novy-Marx and Velikov (2023), demonstrates that PRS-based trading strategies generate substantial risk-adjusted returns that persist even after accounting for well-established risk factors and closely related anomalies.

The key findings reveal that a value-weighted long/short strategy based on PRS delivers an annualized Sharpe ratio of 0.45 gross (0.39 net of transaction costs), indicating strong economic significance even after accounting for implementation frictions. The strategy’s ability to generate monthly abnormal returns of 26 basis points gross relative to the Fama-French five-factor model plus momentum, with a robust t-statistic of 3.22, underscores its statistical reliability. Particularly noteworthy is the signal’s resilience when tested against a comprehensive set of twelve factors, including six closely related payout and issuance-based strategies. The persistence of an 18 basis point monthly alpha (t-statistic = 2.44) in this demanding specification suggests that PRS captures unique information about firm fundamentals not fully reflected in existing factor models or related signals.

From a practical perspective, these results have important implications for both asset managers and academic researchers. The signal’s net Sharpe ratio of 0.39 indicates that PRS-based strategies remain profitable after realistic transaction costs, making them potentially implementable in real-world portfolio management. The distinct information content of PRS, evidenced by its incremental alpha over related strategies, suggests that investors may benefit from incorporating this signal into multi-factor models or as a complement to existing payout and financing-based investment strategies.

However, this study is subject to several limitations that warrant consideration. First, while our analysis follows best practices for anomaly testing, the proliferation of factors in the literature raises concerns about data mining and the multiple testing problem. Second, our sample period and geographic focus may limit the generalizability of findings to other time periods or international markets. Third, while we account for transaction costs, other implementation challenges such as short-sale constraints, market impact for large positions, and capacity constraints may further erode real-world returns.

Future research should explore several promising avenues. International out-of-sample tests would provide valuable evidence on the global applicability of PRS. Time-series analysis of the signal’s performance across different market regimes and economic cycles could illuminate when the strategy performs best and worst. Additionally, investigating the economic mechanisms underlying PRS’s predictive power—whether it reflects risk compensation, behavioral biases, or institutional frictions—would enhance our theoretical understanding. Finally, examining the signal’s effectiveness in different size segments and its interaction with other corporate events such as mergers, acquisitions, or management changes could provide deeper insights into the boundaries of its applicability. As the factor zoo continues to expand, rigorous testing protocols like those employed in this study remain essential for dis-

tinguishing robust return predictors from statistical artifacts.

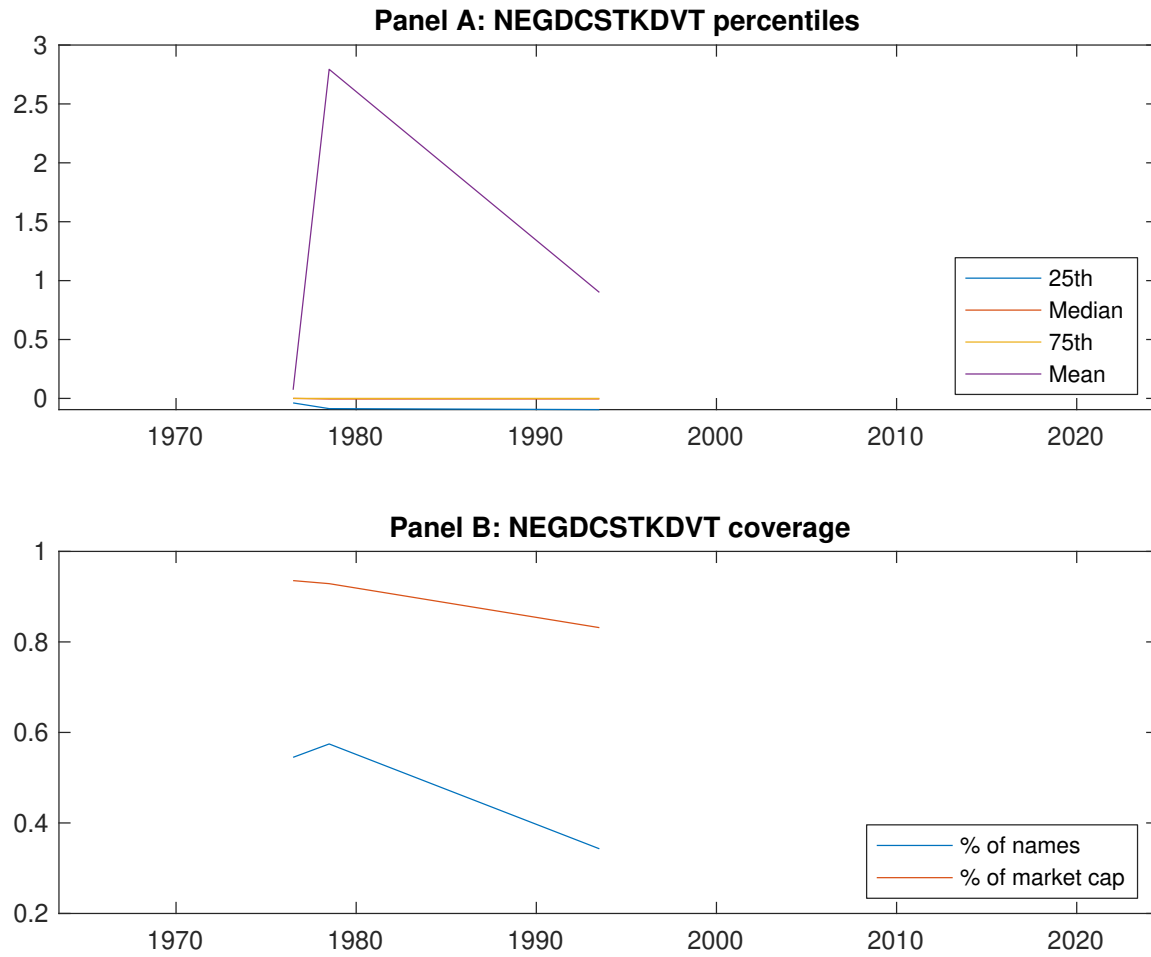


Figure 1: Times series of PRS percentiles and coverage.
This figure plots descriptive statistics for PRS. Panel A shows cross-sectional percentiles of PRS over the sample. Panel B plots the monthly coverage of PRS relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on PRS. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Excess returns and alphas on PRS-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.46 [2.68]	0.58 [3.44]	0.59 [3.65]	0.55 [3.44]	0.73 [4.69]	0.28 [3.51]
α_{CAPM}	-0.11 [-1.70]	0.01 [0.14]	0.04 [0.84]	0.02 [0.33]	0.21 [3.88]	0.32 [4.01]
α_{FF3}	-0.14 [-2.23]	-0.03 [-0.68]	0.02 [0.51]	-0.06 [-1.29]	0.15 [3.10]	0.29 [3.66]
α_{FF4}	-0.14 [-2.19]	-0.02 [-0.31]	0.02 [0.42]	-0.05 [-1.10]	0.15 [2.97]	0.29 [3.55]
α_{FF5}	-0.22 [-3.67]	-0.06 [-1.12]	-0.05 [-1.23]	-0.18 [-3.94]	0.03 [0.73]	0.26 [3.22]
α_{FF6}	-0.22 [-3.55]	-0.04 [-0.79]	-0.05 [-1.15]	-0.16 [-3.51]	0.04 [0.89]	0.26 [3.22]
Panel B: Fama and French (2018) 6-factor model loadings for PRS-sorted portfolios						
β_{MKT}	0.99 [67.40]	1.00 [82.92]	0.99 [93.70]	1.00 [90.60]	0.97 [85.24]	-0.02 [-1.09]
β_{SMB}	-0.00 [-0.04]	-0.03 [-1.46]	-0.06 [-4.23]	-0.10 [-6.54]	-0.06 [-3.54]	-0.06 [-2.05]
β_{HML}	0.11 [3.93]	0.13 [5.58]	0.03 [1.62]	0.14 [6.50]	0.07 [3.41]	-0.04 [-0.98]
β_{RMW}	0.27 [9.29]	0.09 [3.90]	0.18 [8.67]	0.21 [9.67]	0.21 [9.30]	-0.06 [-1.59]
β_{CMA}	-0.05 [-1.11]	-0.04 [-1.10]	0.07 [2.41]	0.25 [7.97]	0.24 [7.52]	0.29 [5.26]
β_{UMD}	-0.01 [-0.39]	-0.02 [-1.92]	-0.00 [-0.40]	-0.03 [-2.30]	-0.01 [-0.98]	-0.01 [-0.28]
Panel C: Average number of firms (n) and market capitalization (me)						
n	405	354	317	352	391	
me (\$10 ⁶)	1150	1200	1690	1678	1956	

Table 2: Robustness to sorting methodology & trading costs

This table evaluates the robustness of the choices made in the PRS strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016b) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.28 [3.51]	0.32 [4.01]	0.29 [3.66]	0.29 [3.55]	0.26 [3.22]	0.26 [3.22]
Quintile	NYSE	EW	0.33 [6.03]	0.39 [7.25]	0.33 [6.55]	0.28 [5.45]	0.24 [4.87]	0.20 [4.08]
Quintile	Name	VW	0.28 [3.38]	0.31 [3.82]	0.28 [3.43]	0.28 [3.30]	0.25 [3.04]	0.25 [3.01]
Quintile	Cap	VW	0.23 [3.03]	0.27 [3.48]	0.26 [3.37]	0.26 [3.24]	0.23 [3.02]	0.24 [2.99]
Decile	NYSE	VW	0.35 [3.81]	0.38 [4.14]	0.32 [3.49]	0.29 [3.17]	0.32 [3.47]	0.30 [3.24]
Panel B: Net Returns and Novy-Marx and Velikov (2016b) generalized alphas								
Portfolios	Breaks	Weights	r_{net}^e	α_{CAPM}^*	α_{FF3}^*	α_{FF4}^*	α_{FF5}^*	α_{FF6}^*
Quintile	NYSE	VW	0.24 [3.06]	0.28 [3.53]	0.25 [3.16]	0.25 [3.12]	0.23 [2.94]	0.23 [2.94]
Quintile	NYSE	EW	0.19 [3.31]	0.26 [4.60]	0.21 [3.84]	0.18 [3.33]	0.12 [2.37]	0.11 [2.05]
Quintile	Name	VW	0.24 [2.94]	0.27 [3.31]	0.24 [2.91]	0.23 [2.86]	0.22 [2.73]	0.22 [2.72]
Quintile	Cap	VW	0.20 [2.59]	0.23 [3.03]	0.22 [2.86]	0.22 [2.81]	0.21 [2.71]	0.21 [2.71]
Decile	NYSE	VW	0.31 [3.34]	0.33 [3.59]	0.27 [2.99]	0.26 [2.83]	0.27 [3.03]	0.27 [2.94]

Table 3: Conditional sort on size and PRS

This table presents results for conditional double sorts on size and PRS. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on PRS. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high PRS and short stocks with low PRS. Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 1963Q6 to 2024Q6.

Panel A: portfolio average returns and time-series regression results												
Size quintiles	PRS Quintiles					PRS Strategies						
		(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
	(1)	0.63 [2.75]	0.82 [3.61]	0.88 [3.19]	0.93 [4.36]	0.95 [4.55]	0.35 [3.80]	0.40 [4.31]	0.35 [3.90]	0.32 [3.51]	0.25 [2.72]	0.23 [2.48]
	(2)	0.65 [3.00]	0.78 [3.69]	0.83 [3.99]	0.97 [4.61]	0.85 [4.21]	0.20 [2.20]	0.24 [2.74]	0.19 [2.15]	0.20 [2.25]	0.13 [1.44]	0.15 [1.62]
	(3)	0.64 [3.31]	0.65 [3.26]	0.79 [4.04]	0.75 [3.98]	0.93 [4.92]	0.29 [3.41]	0.31 [3.69]	0.26 [3.08]	0.24 [2.83]	0.21 [2.52]	0.21 [2.38]
	(4)	0.52 [2.75]	0.66 [3.56]	0.74 [3.93]	0.66 [3.71]	0.81 [4.54]	0.29 [3.85]	0.33 [4.44]	0.28 [3.82]	0.29 [3.88]	0.19 [2.64]	0.21 [2.85]
	(5)	0.43 [2.56]	0.53 [3.21]	0.49 [3.04]	0.50 [3.09]	0.70 [4.50]	0.27 [2.93]	0.30 [3.29]	0.28 [3.06]	0.28 [2.94]	0.27 [2.90]	0.27 [2.86]
Panel B: Portfolio average number of firms and market capitalization												
Size quintiles	PRS Quintiles					PRS Quintiles						
		Average n					Average market capitalization (\$10 ⁶)					
		(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)	
	(1)	151	151	151	151	148	14	14	15	12	12	
	(2)	63	62	63	63	62	28	28	28	28	27	
	(3)	53	53	52	52	52	56	55	57	57	57	
	(4)	49	49	49	49	49	134	137	140	139	142	
(5)	52	52	52	52	52	1048	1249	1395	1249	1554		

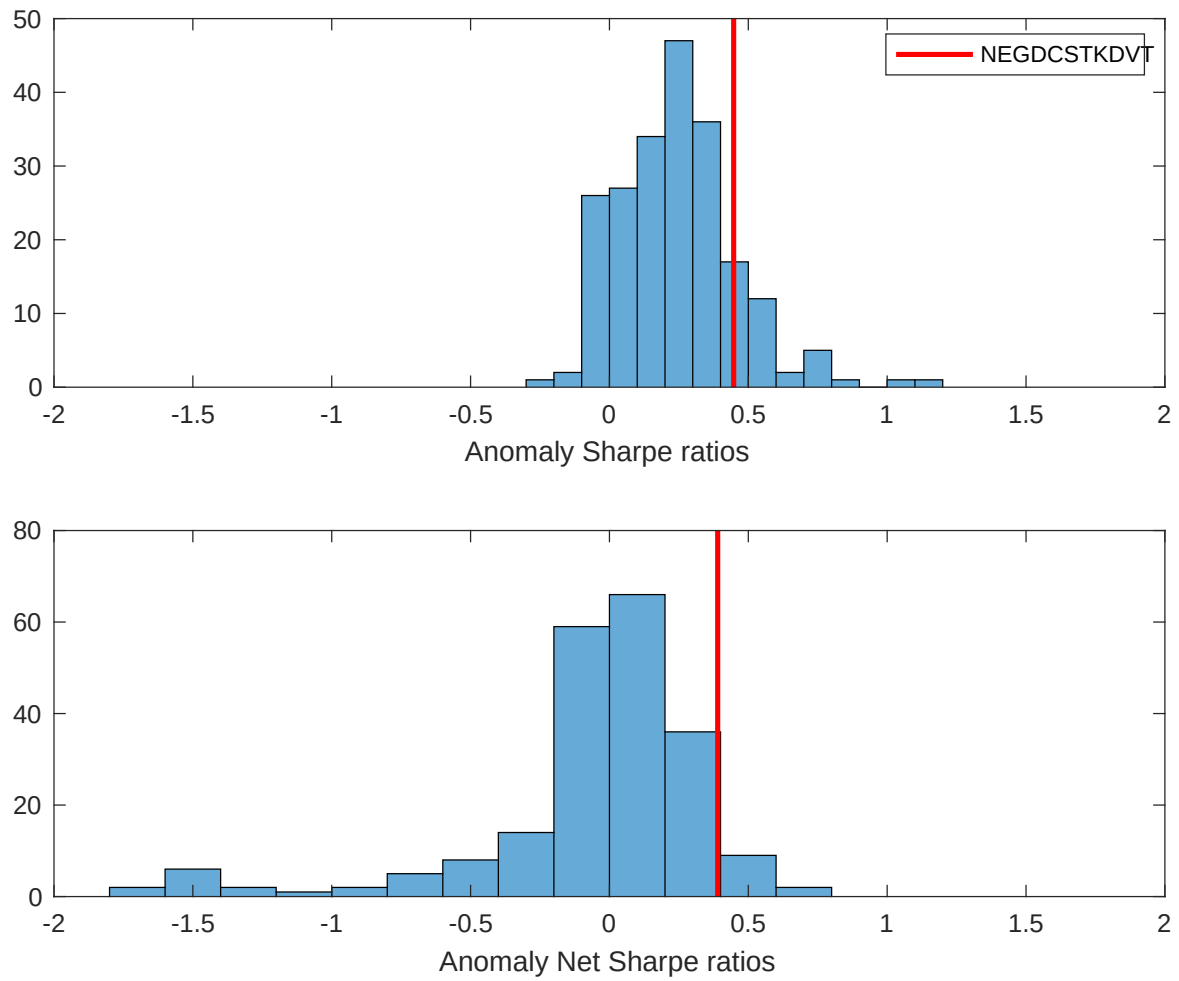


Figure 2: Distribution of Sharpe ratios.

This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the PRS with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

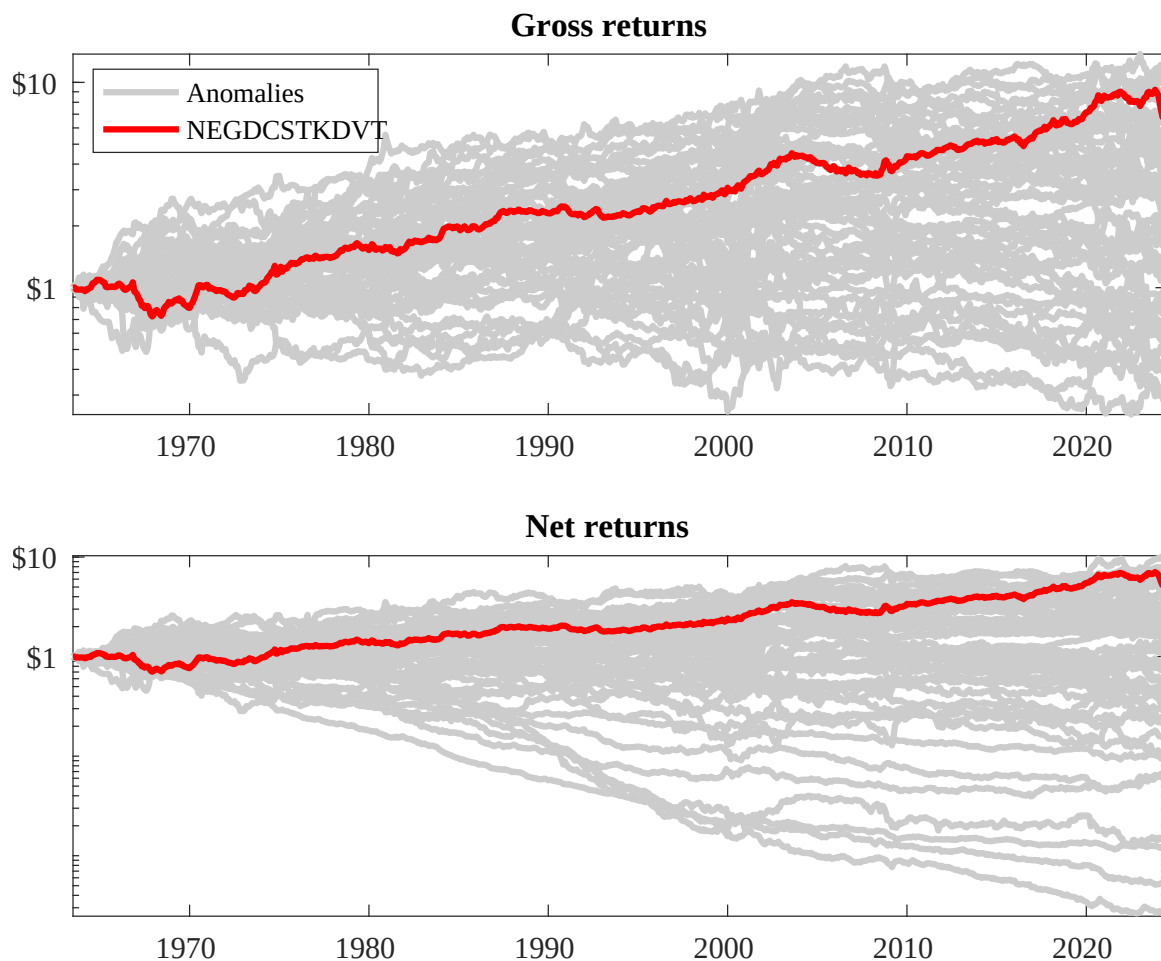
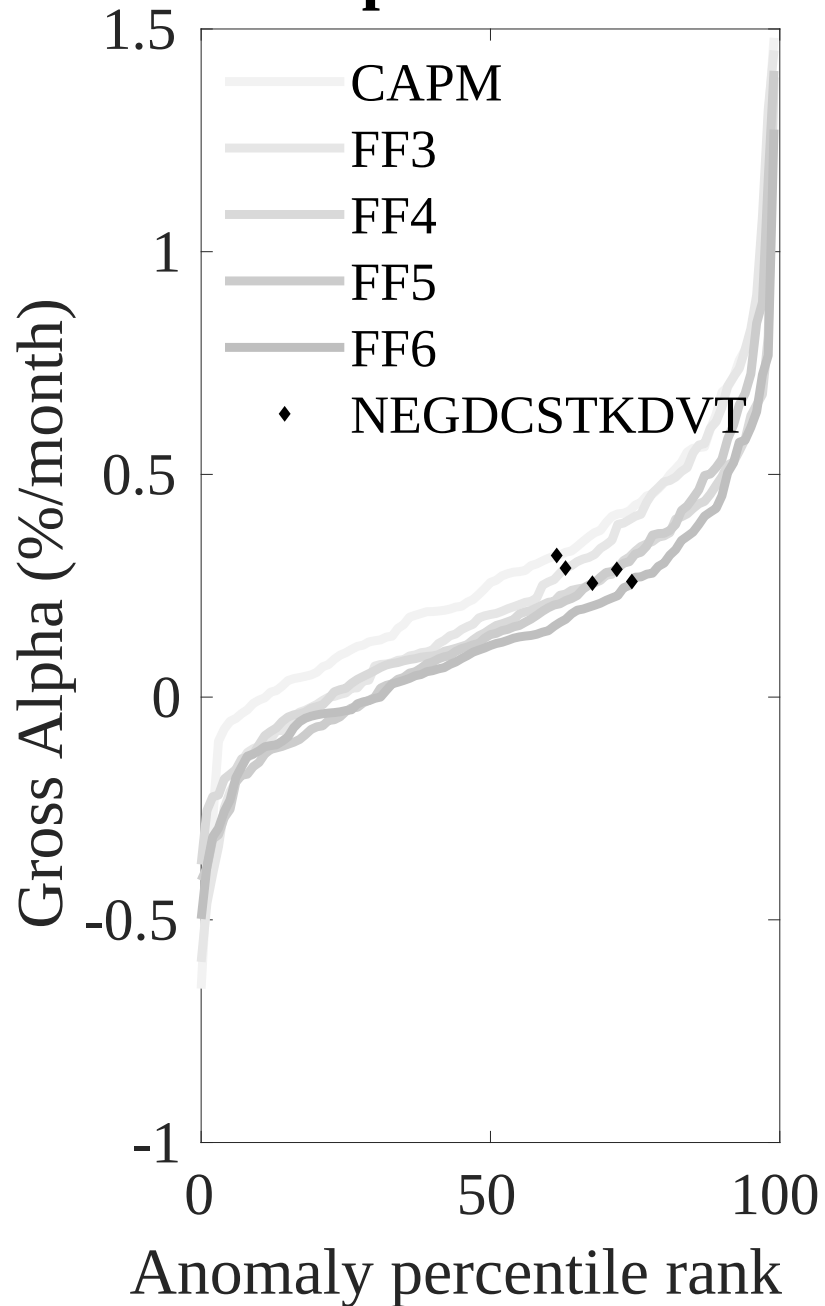


Figure 3: Dollar invested.

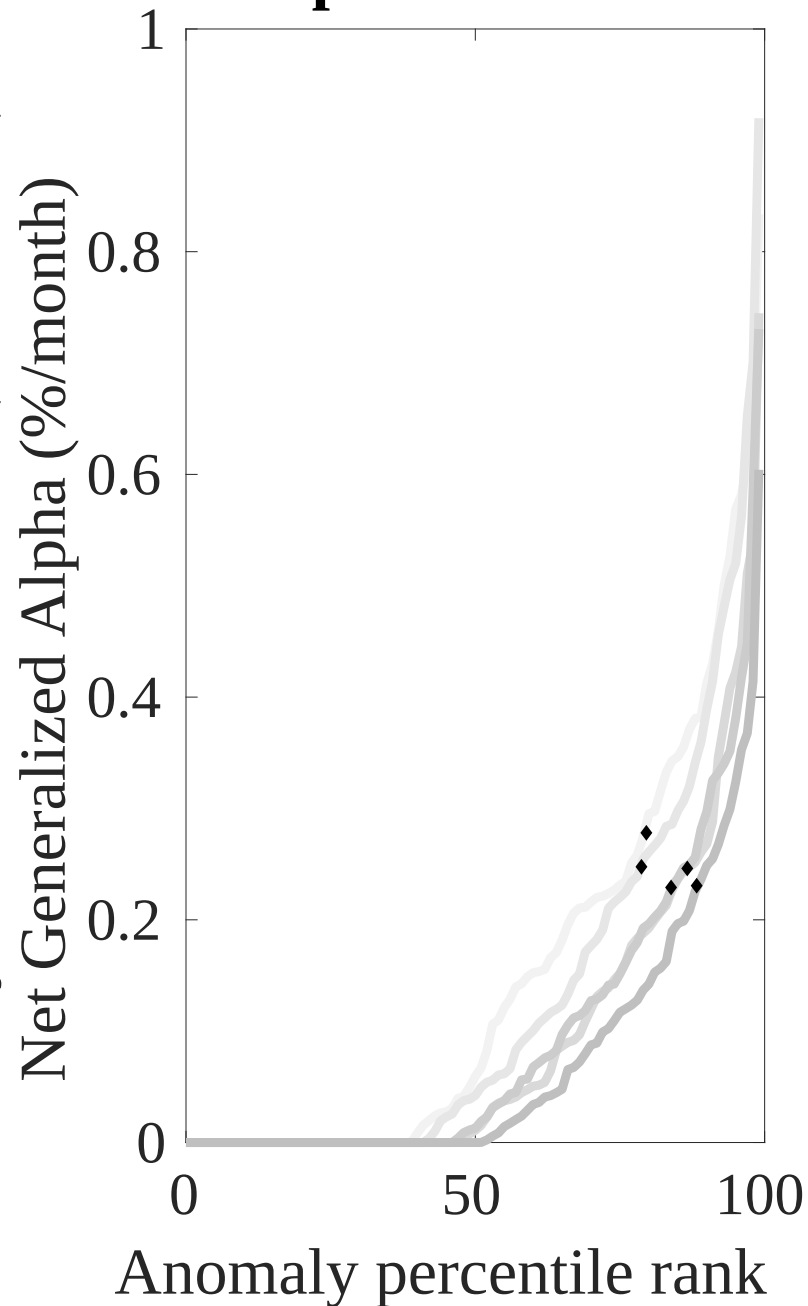
This figure plots the growth of a \$1 invested in 212 anomaly trading strategies (gray lines), and compares those with the PRS trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strategy returns.

Gross Alpha Percentiles



Novy-Marx and Velikov (RFS, 2016)

Net Alpha Percentiles



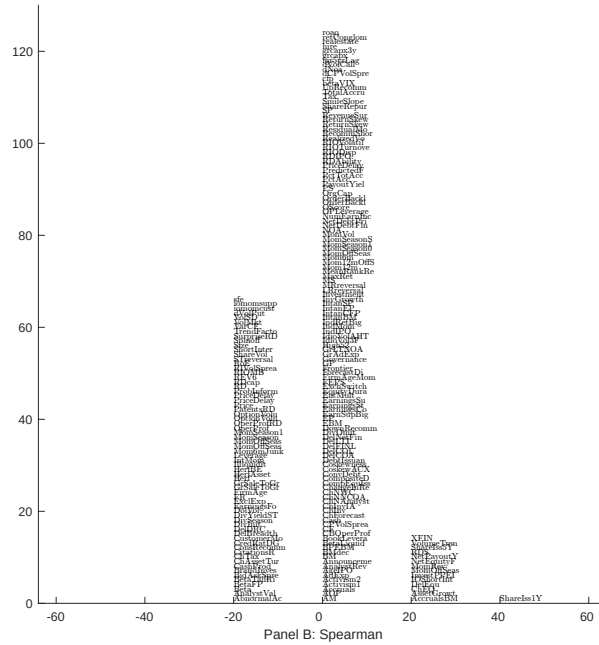
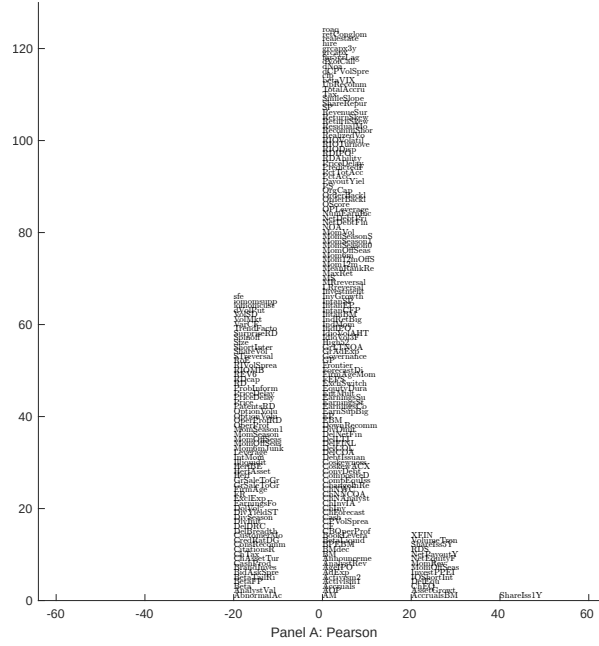


Figure 5: Distribution of correlations.

This figure plots a name histogram of correlations of 205 filtered anomaly signals with PRS. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

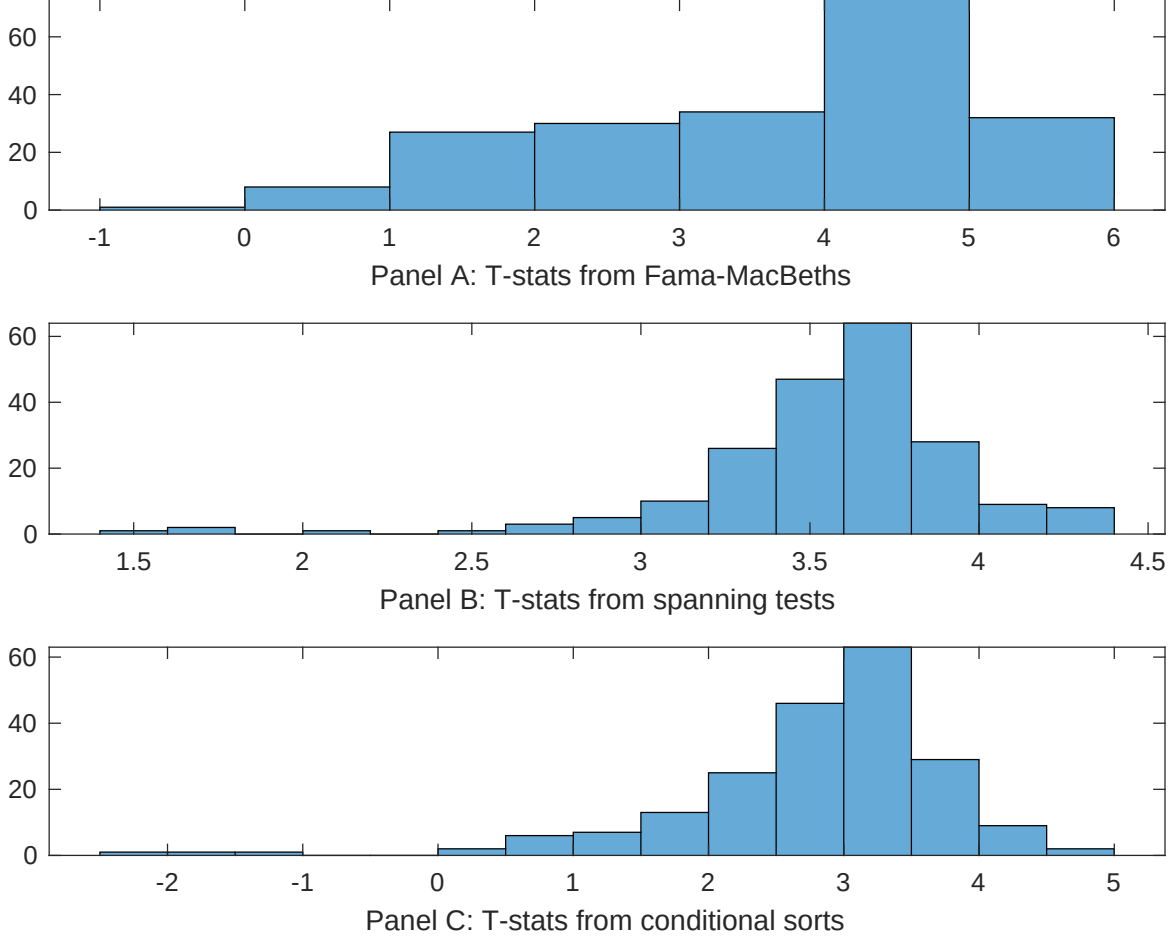


Figure 7: Distribution of t-stats on conditioning strategies

This figure plots histograms of t-statistics for predictability tests of PRS conditioning on each of the 205 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{PRS} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{PRS} PRS_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 205 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{PRS,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 205 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 205 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on PRS. Stocks are finally grouped into five PRS portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted PRS trading strategies conditioned on each of the 205 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on PRS. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{PRS} PRS_{i,t} + \sum_{k=1}^s \beta_{X_k} X_{i,t}^k + \epsilon_{i,t}$. The six most closely related anomalies, X , are Growth in book equity, Net Payout Yield, Share issuance (1 year), Share issuance (5 year), Change in equity to assets, Asset growth. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.15 [6.41]	0.11 [5.53]	0.12 [6.45]	0.12 [6.65]	0.12 [6.28]	0.12 [6.66]	0.12 [3.87]
PRS	0.48 [3.24]	0.42 [1.74]	0.49 [3.43]	0.39 [2.60]	0.56 [3.68]	0.54 [3.56]	0.25 [1.03]
Anomaly 1	0.27 [2.10]						-0.22 [-0.09]
Anomaly 2		0.23 [2.06]					0.11 [1.00]
Anomaly 3			0.13 [5.51]				0.45 [1.49]
Anomaly 4				0.24 [3.98]			0.23 [3.71]
Anomaly 5					0.10 [2.50]		-0.11 [-0.17]
Anomaly 6						0.70 [5.48]	0.55 [4.28]
# months	720	715	715	715	720	720	715
$\bar{R}^2(\%)$	1	1	1	1	1	0	0

Table 5: Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the PRS trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{PRS} = \alpha + \sum_{k=1}^6 \beta_{X_k} r_t^{X_k} + \sum_{j=1}^6 \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor. The six most closely related anomalies, X , are Growth in book equity, Net Payout Yield, Share issuance (1 year), Share issuance (5 year), Change in equity to assets, Asset growth. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.24 [3.08]	0.29 [3.67]	0.25 [3.16]	0.18 [2.32]	0.25 [3.14]	0.25 [3.09]	0.18 [2.44]
Anomaly 1	23.29 [5.44]						37.22 [6.23]
Anomaly 2		16.53 [5.45]					8.35 [2.55]
Anomaly 3			24.31 [6.16]				7.39 [1.57]
Anomaly 4				31.63 [7.70]			20.53 [4.41]
Anomaly 5					2.36 [0.59]		-18.08 [-3.29]
Anomaly 6						-14.25 [-2.99]	-24.52 [-4.77]
mkt	-1.21 [-0.64]	0.83 [0.43]	-1.13 [-0.61]	-0.33 [-0.18]	-2.01 [-1.05]	-1.99 [-1.04]	2.07 [1.13]
smb	-5.37 [-1.97]	-1.87 [-0.68]	-3.61 [-1.34]	-2.41 [-0.90]	-4.62 [-1.66]	-3.45 [-1.24]	-0.37 [-0.14]
hml	-5.24 [-1.42]	-8.36 [-2.19]	-2.53 [-0.70]	0.54 [0.15]	-2.66 [-0.71]	-1.83 [-0.49]	-5.39 [-1.48]
rmw	-4.95 [-1.34]	-15.22 [-3.75]	-18.75 [-4.45]	-16.95 [-4.38]	-5.83 [-1.55]	-5.84 [-1.57]	-21.61 [-4.97]
cma	5.51 [0.81]	13.61 [2.32]	12.41 [2.14]	12.25 [2.21]	25.70 [3.79]	45.28 [5.72]	18.20 [2.32]
umd	-0.77 [-0.41]	0.99 [0.53]	-1.43 [-0.77]	-2.39 [-1.30]	-0.28 [-0.15]	-0.76 [-0.40]	-3.18 [-1.77]
# months	732	728	728	728	732	732	728
$\bar{R}^2(\%)$	11	11	12	14	7	8	21

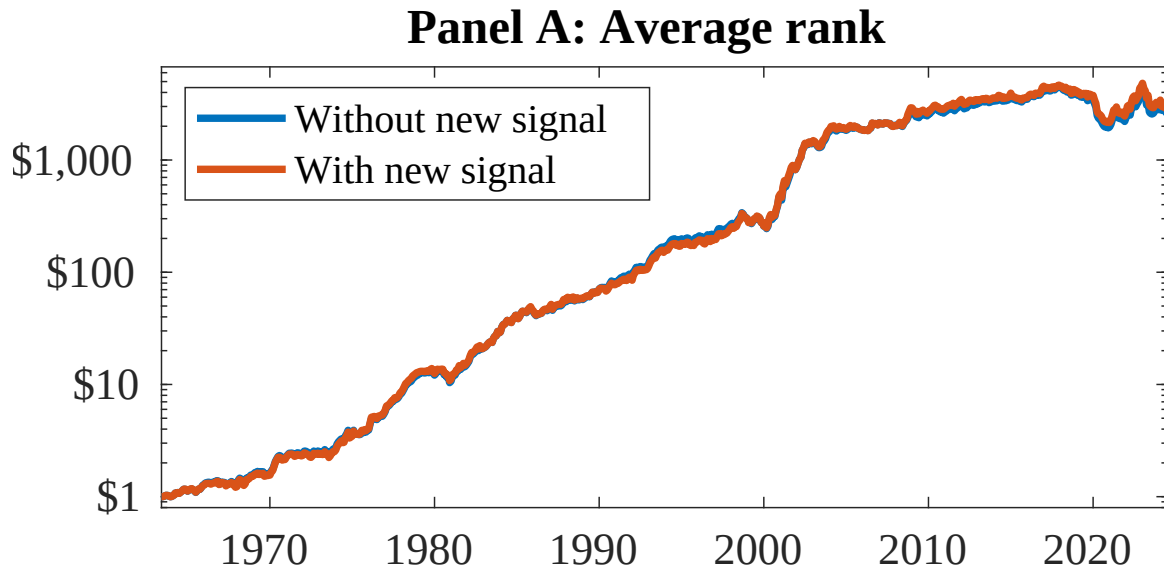


Figure 8: Combination strategy performance

This figure plots the growth of a \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). In all panels, the blue solid lines indicate combination trading strategies that utilize 154 anomalies. The red solid lines indicate combination trading strategies that utilize the 154 anomalies as well as PRS. Panel A shows results using "Average rank" as the combination method. See Section 7 for details on the combination methods.

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